Google Play Store Dataset Analysis

Introduction

This project analyzes a Google Play Store dataset containing information about mobile applications: category, rating, reviews, installs, price, size, and type (free or paid). The goal is to identify patterns and key factors that influence an application's success.

Business Context

In a market where the number of available applications exceeds millions, understanding what characteristics contribute to an app's success is essential for guiding product and monetization strategies.

General Objective

Determine the key factors that influence an application's success on Google Play Store, measured in terms of installs and user ratings.

Specific Objectives

1. Identify the categories with the highest number of installs and best ratings.
2. Analyze the influence of price and app type on user acceptance.
3. Explore the relationship between reviews, installs, and satisfaction.
4. Detect market opportunities in categories with high potential and low competition.

Data Cleaning Process

Before conducting the analysis, the raw dataset was cleaned and transformed using PostgreSQL. The following steps were applied to ensure data quality and consistency.

Step 1: Initial Exploration — Reviewed table structure, column names, and data types using information_schema.columns. Previewed key columns to identify potential issues.

Step 2: Problem Detection — Used regex patterns to detect non-numeric values in installs, reviews, size, and price. Checked for null values, invalid ratings (outside 1-5 range), and duplicate records.

Step 3: Duplicate Removal — Removed 1,170 duplicate entries using the ROW_NUMBER() window function, keeping only the row with the most reviews per app.

Step 4: Size Column Cleaning — Created a new numeric column (size_mb). Converted 'M' and 'k' suffixes to megabytes using CASE WHEN. Set 'Varies with device' to NULL.

Step 5: Data Type Conversion — Converted reviews, installs, and price from text to NUMERIC. Converted last_updated from text to DATE using TO_DATE().

Step 6: Categorical Validation — Verified consistency of type (Free/Paid), category (33 unique values), and content_rating (6 unique values). No inconsistencies found.

Full cleaning queries available in: cleaning_google_play.sql

Analysis Questions & Findings

The following six questions were defined to guide the analysis. Each question was answered using SQL queries and targets a specific analytical skill.

1. Which categories have the most installs and best average rating?

Skill: Grouping and sorting

Finding: GAME leads with 590M total installs and a 4.24 average rating. The top 5 categories by installs are: Game, Communication, Tools, Social, and Family.

See Question 1 in analysis_google_play.sql

2. Do paid apps have better ratings than free apps?

Skill: Segment comparison

Finding: Yes, paid apps have a slightly higher average rating (4.26) compared to free apps (4.17). However, the difference is minimal, suggesting that price alone is not a strong indicator of quality.

See Question 2 in analysis_google_play.sql

3. What is the price range with the highest acceptance in terms of installs and rating?

Skill: Segmentation and business logic

Finding: The $0–$5 range offers the best balance: highest rating among paid apps (4.28) and the most installs. Apps priced above $10 have the lowest rating (4.13) and nearly zero installs.

See Question 3 in analysis_google_play.sql

4. Is there a relationship between the number of reviews and the rating?

Skill: Correlation analysis

Finding: Strong positive relationship. High-rated apps (4–5 stars) average 338,930 reviews, while low-rated apps (1–2 stars) average only 180. Higher user satisfaction drives significantly more engagement.

See Question 4 in analysis_google_play.sql

5. Which categories are saturated and which represent a growth opportunity?

Skill: Market vision

Finding: 3 saturated categories were identified: Family (1,651 apps), Game (898), and Tools (720). 5 opportunity categories were found: Communication, Photography, Social, Video Players, and Entertainment — all with fewer than 400 apps but above-average installs (>255,500) and high ratings (>4.0).

See Question 6 in analysis_google_play.sql

Tools Used

- PostgreSQL — Data cleaning, transformation, and analytical queries
- Power BI — Dashboard and data visualization

Project Files

- cleaning_google_play.sql — Full data cleaning process with step-by-step comments
- analysis_google_play.sql — Analytical queries for all 6 questions
- apps_clean.csv — Clean dataset exported for visualization
- dashboard.pbix — Power BI interactive dashboard

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